

Polynomial rings



Recall a field is a commutative ring in

which all nonzero element is a unit.

In particular, any field is a domain.

Def 1: Let K be a field. We say

$$K[X] = \left\{ a_n X^n + a_{n-1} X^{n+1} + \dots + a_1 X + a_0 : \right. \\ \left. a_0, \dots, a_n \in K, n \in \mathbb{N} \right\}.$$

is the ring of polynomials with coefficients

in K . The addition and multiplication

is defined as (write $a_n X^n + \dots + a_1 X + a_0$

$$\text{as } \sum_{i=0}^n a_i X^i, \text{ and } a_k = 0 \ \forall k > n)$$

$$+ : k[x] \times k[x] \rightarrow k[x].$$

$$\left(\sum_{i=0}^n a_i x^i, \sum_{j=0}^m b_j x^j \right) = \sum_{i=0}^n a_i x^i + \sum_{j=0}^m b_j x^j$$

$$= \sum_{d=0}^{\max(m,n)} (a_d + b_d) x^d$$

$$\cdot : k[x] \times k[x] \rightarrow k[x].$$

$$\left(\sum_{i=0}^n a_i x^i, \sum_{j=0}^m b_j x^j \right) \mapsto \left(\sum_{i=0}^n a_i x^i \right) \left(\sum_{j=0}^m b_j x^j \right)$$

$$= \sum_{d=0}^{m+n} \left(\sum_{i+j=d} a_i b_j \right) x^d$$

We call $a_0 \in k \subseteq k[x]$ the constant polynomials.

$$\text{eg. } (12x^2 + 3x + 5) + (9x^3 - 1)$$

$$= 9x^3 + 12x^2 + 3x + 4$$

$$(x-1) (x^2 + x + 1)$$

$$= x^3 + 0x^2 + 0x - 1$$

$$\text{def 2: let } f(x) = a_n x^n + \dots + a_1 x + a_0$$

be in $K[x]$. with $a_n \neq 0$, and $n \geq 0$.

We say $\deg f(x) = n$. In particular,

we say all nonzero constant polynomials

are of $\deg 0$. and $\deg 0 = -\infty$.

$$\text{eg. } \deg x^4 + x^{17} - 3 + 7x^{10} = 17$$

lemma 3: let $f(x), g(x) \in K[x]$ such

that $\deg f(x) = n$. $\deg g(x) = m$.

Then. $\deg f(x)g(x) = m+n$.

proof. Suppose $n = -\infty$ or $m = -\infty$

Then. $f(x) = 0$ or $g(x) = 0$.

which implies $f(x)g(x) = 0$ and is of $\deg -\infty$

Suppose both $n, m \geq 0$

Then. write $f(x) = a_n x^n + \dots + a_1 x + a_0$

$$g(x) = b_m x^m + \dots + b_1 x + b_0$$

for $a_n, b_m \neq 0$,

Then. $f(x)g(x) = a_n b_m x^{n+m} + \text{lower deg terms.}$

Note $a_n b_m \neq 0$ as k is a field

So $\deg f(x)g(x) = n+m.$

Def 4: Let $f(x) = a_n x^n + \dots + a_1 x + a_0 \neq 0$

be in $k[x]$. We say the leading term

of $f(x)$ is $LT(f(x)) = a_m x^m$ where

$m = \max \{d : a_d \neq 0\}$. We

then call the coefficient of the leading

term of $f(x)$ as the leading coefficient:

which we denote by $LC(f(x))$.

Lemma 5. Let $f(x), g(x) \in k[x]$.

$$\text{Then } LT(f(x)g(x)) = LT(f(x)) \cdot LT(g(x))$$

$$\text{and } LC(f(x)g(x)) = LC(f(x)) \cdot LC(g(x))$$

Lemma 6: Let k be a field.

Then $k[x]$ is a domain.

proof. Suppose $0 \neq f(x)$ is a zero-divisor.

Then $\exists 0 \neq g(x) \in k[x]$ such that

$$f(x)g(x) = 0 \quad \text{which implies}$$

$$LC(f(x)) \cdot LC(g(x)) = 0 \quad \text{in } k.$$

It follows $LC(f(x)) = 0$ or $LC(g(x)) = 0$.

Cor 7: let k be a field. Then.

$0 \neq f(x) \in k[x]$ is a unit iff $\deg f = 0$.

proof. We note that

$0 \neq f(x) \in k[x]$ is a unit

iff $\exists 0 \neq g(x) \in k[x]$ s.t

$$f(x)g(x) = 1$$

iff $\deg f(x) + \deg g(x) = 0$

iff $\deg f(x) = \deg g(x) = 0$

Def 8. Let $f(x), g(x) \in k[x]$.

We say $f(x)$ divides $g(x)$ if

$$\exists h(x) \in k[x]. \text{ s.t. } h(x)f(x) = g(x)$$

We say $f(x) \in k[x]$ is irreducible

if $f(x) \neq 0$ is not a unit

and whenever $f(x) = g(x)h(x)$

for $g(x), h(x) \in k[x]$. $\deg g(x) = 0$

or $\deg h(x) = 0$.

eg. $x^2 + 1$ is irreducible in $\mathbb{R}[x]$.

factors as $(x+i)(x-i)$ in $\mathbb{C}[x]$

Lemma 9: Let K be a field. Then.

any polynomial $f(x)$ of positive degree.

is either irreducible or factors as

a product of irreducibles.

Lemma 10. Let K be field. The polynomial

ring $K[x]$ has infinitely many

irreducibles.

Exercise :

1. Is $\mathbb{Z}_6[X]$ a domain ?

2. Does $\deg(f(x)g(x)) = \deg f(x) + \deg g(x)$
holds. ?

3.